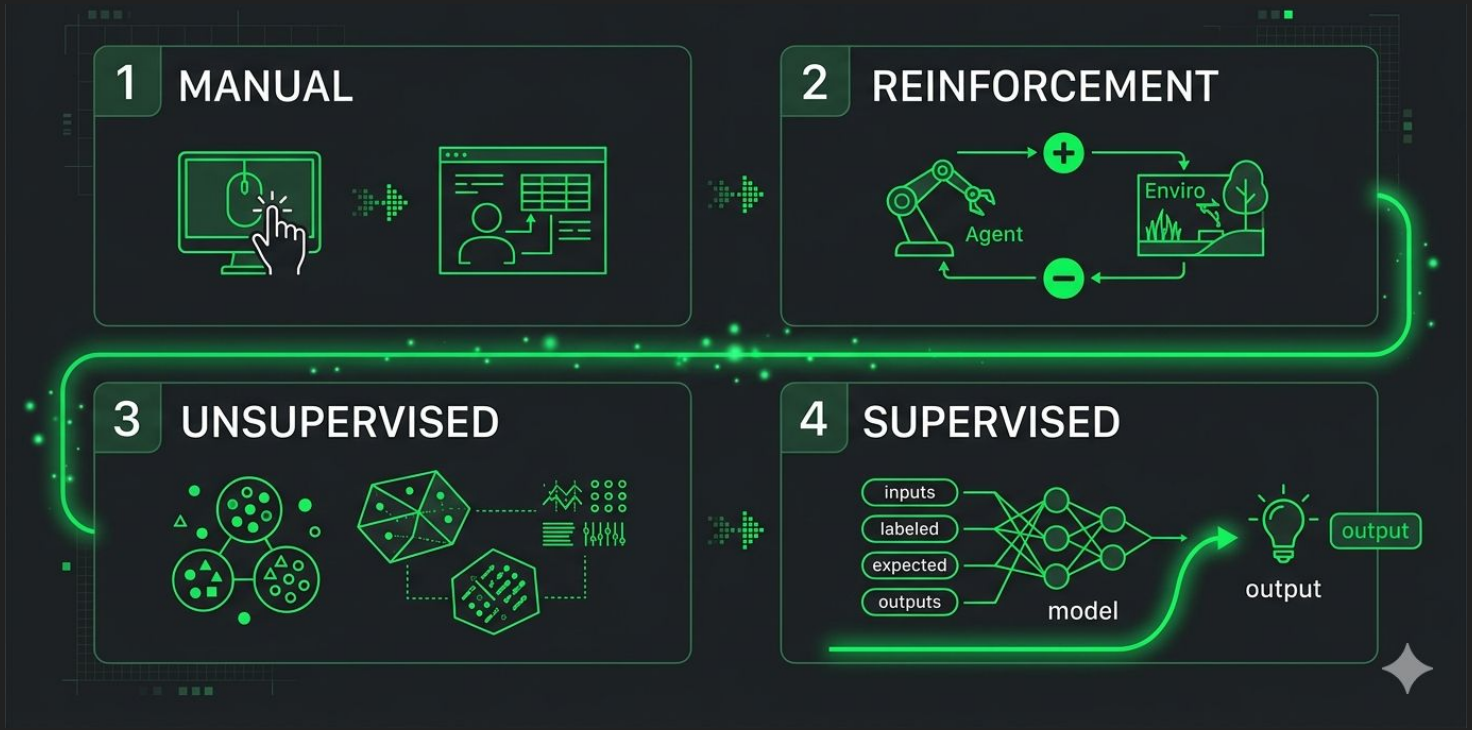
AI Evolution

A Multi-Phase Workshop for Machine Learning

The Evolution Roadmap



Real World Use Case

Getting Started

https://github.com/joshm21/ug-workshop1

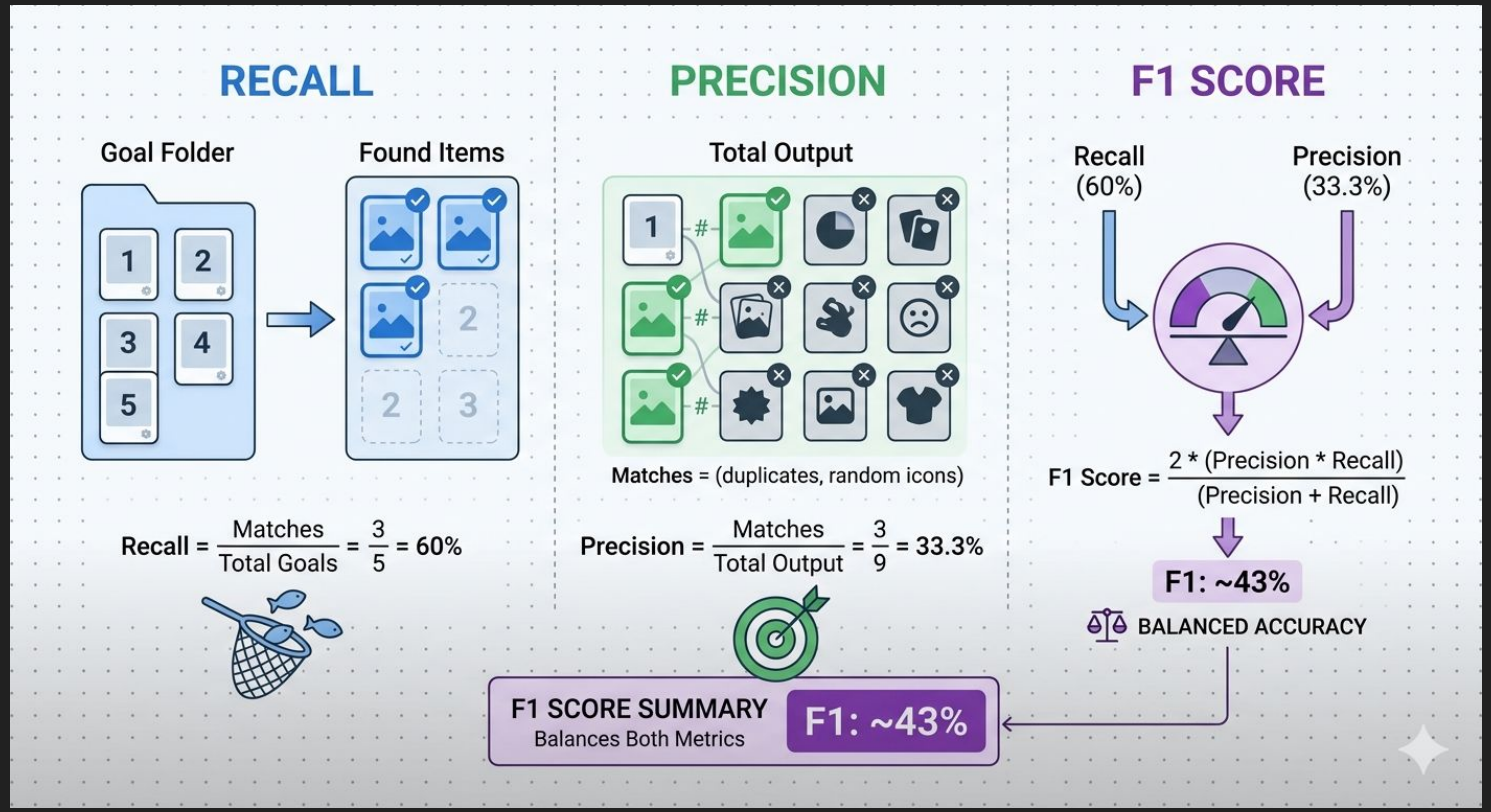
Project Structure

```
config.py        # the main file you will edit
run.py           # run the pipeline with '%run run.py'
evolve.py        # edit during Phase 2
examples/        # starting examples for each phase
core/            # provided libraries; do not edit
data/video-id/           # data files related to each video
    video.mp4            # the source video file
    goal/                # the expected image outputs
    artifacts/run-id/    # generated image / data files
```

The Pipeline Pseudocode

```
For video in CONFIG.VIDEOS:
    For ts in CONFIG.TIMESTAMPS:
        For step in FRAME_PIPELINE:
            step.run()
    video_image_crops.deduplicate()
    video.grade()
all_videos.grade()
```

The Pipeline Score



Phase 1: The Human Baseline

Check the examples/Phase 1 folder

Note the provided libraries in /core

- prepare, detect, score, plot, filter
- None of these need to be modified during the workshop

Phase 1: Challenge

Goal: Perfectly extract every picture card

Task: adjust ‘FRAME_PIPELINE’ by adding, deleting, reordering, or modifying functions using the prepare, detect, score, plot and filter libraries in core/

What accuracy can you achieve across all videos?

The Brittleness Problem

Phase 2: Genetic Algorithms (Reinforcement)

```
# pre-process
prepare.to_grayscale,
partial(prepare.to_blurred, ksize=3),
partial(prepare.at_canny_edges, low=50, high=200),
partial(prepare.do_dilation, kernel_size=3, iterations=1),

# detect, score, visualize
partial(detect.find_quads, min_area=50, epsilon=0.03),
```

1 function = 1 gene

Genome = group of genes (eg group of functions = 1 config)

Population = group of genomes (eg group of configs)

Phase 2: Genetic Algorithms (Reinforcement)

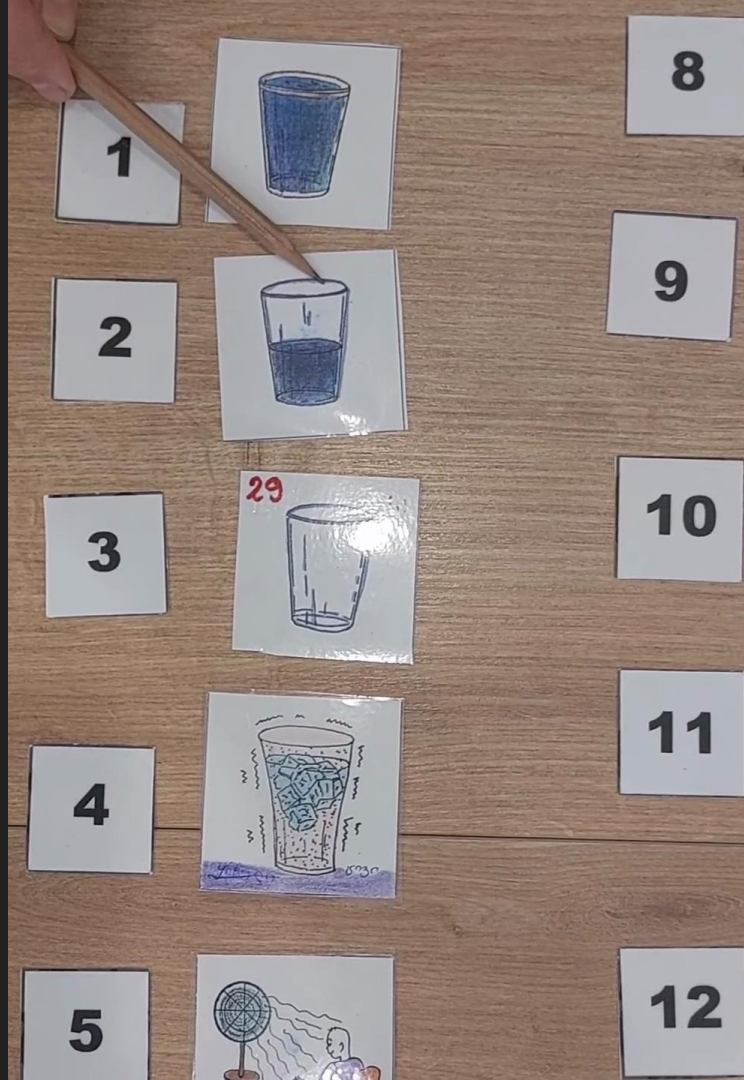
Expected =

full + # covered + # cutoff

Is there a better function?

Fitness =

100 - abs(expected - detected)*10



Phase 2: Genetic Algorithms (Reinforcement)

Crossover =

swap genes (functions) between best genomes (configs)

Mutate =

add, delete, or modify genes (functions and their args)

Repeat for several generations, save best results

Phase 2: Challenge

Goal: Achieve the highest fitness across the training data

Task: adjust evolve.py vars

Bonus: Can you define a better expected_count function?

The Training Data Problem

Phase 3: K-Means Clustering (Unsupervised)

https://www.geeksforgeeks.org/python/k-means-clustering-with-scipy/
GIF Animation: https://en.wikipedia.org/wiki/K-means_clustering#/media/File:K-means_convergence.gif

Phase 3: Challenge

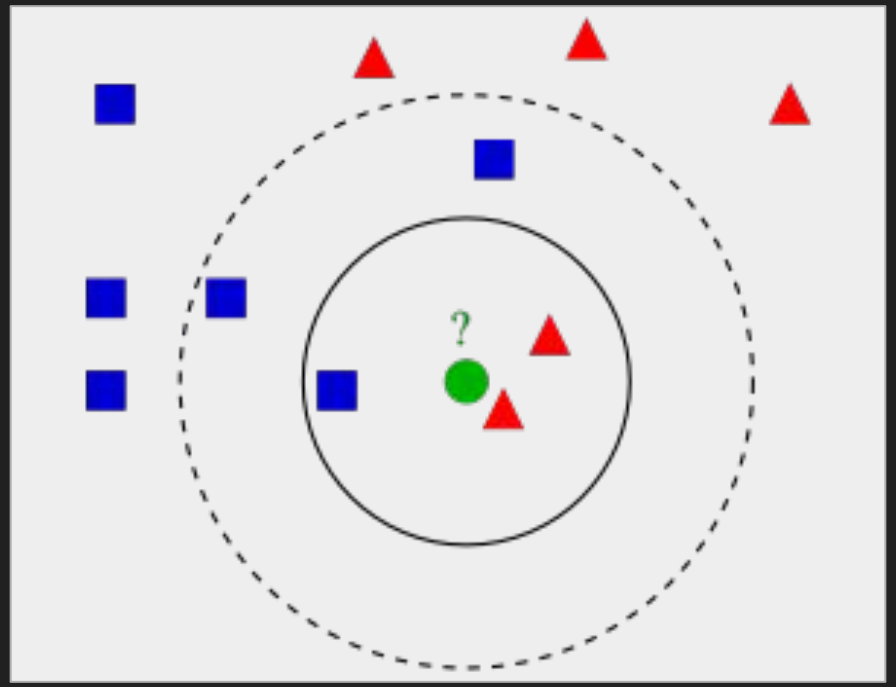
Goal: Perfectly extract every picture card; don’t use any filtering steps

Task: add a cluster.k_means step to the pipeline

What accuracy can you achieve across all videos?

Cluster Selection Problem

Phase 4: k Nearest Neighbors (Supervised)



https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Phase 4: Challenge

Goal: Perfectly extract every picture card using your custom training data

Task: Add a ‘label’ column to your scores csv (1 = Picture, 2 = Number, 3 = Other) and add a classify.knn step to the pipeline

What accuracy can you achieve across all videos?

Summary

Feature	Reinforcement Learning	Unsupervised Learning	Supervised Learning
Core Goal	Learn through trial and error to maximize a reward	Discover hidden patterns or structures in data	Predict an output based on labeled input data
Data Type	Dynamic environment/feedback loop	Unlabeled data (no ground truth)	Labeled data (input-output pairs)
Workshop Phase	Phase 2: Genetic Algorithms	Phase 3: K-Means Clustering	Phase 4: k-Nearest Neighbors
Key Metric	Fitness Score	Clustering Quality	Accuracy/Confidence
Analogy	A dog learning a trick via treats	Sorting a bucket of mixed Legos	A student learning with an answer key